OpenGL Assignment 1: Setup and Environment Configuration

This repository contains the environment configuration and source code for Assignment 1, specifically focusing on the integration of the FreeGLUT library within Visual Studio 2026.

Setup and Installation Steps (Part A)

To fulfill the requirement for a "Short explanation of setup steps," the following configuration was implemented to ensure the project remains portable and clonable:

1. **Library Selection & Compatibility**:
   - Utilized **FreeGLUT** to handle window management and input callbacks.
   - **MSVC Integration**: Switched from MinGW static libraries (`.a` files) to **MSVC binaries** (`.lib`) to ensure compatibility with the Visual Studio 2026 compiler and linker.
   - Binaries were sourced from Martin Payne's FreeGLUT Windows Binaries.
2. **Portable Project Structure**:
   - To ensure the project runs immediately upon cloning, a local dependency structure was established:
     - `/include/GL` : Contains OpenGL header files.
     - `/lib` : Contains the 64-bit static library (`freeglut.lib`).
     - `/bin` : Contains the runtime dynamic link library (`freeglut.dll`).
3. **IDE Configuration (Visual Studio 2026)**:
   - **Include Directories**: Configured to `$(ProjectDir)include` using macros to avoid hardcoded absolute paths.
   - **Library Directories**: Configured to `$(ProjectDir)lib` .
   - **Linker Input**: Explicitly linked `freeglut.lib` , `opengl32.lib` , and `glu32.lib` to handle the core OpenGL and Utility functions.
   - **Runtime Deployment**: Placed `freeglut.dll` in the project root alongside `main.cpp` to ensure the executable can locate the library at runtime.

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